Image Geometrical Transformations

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Image courtesy: https://www.etsy.com/in-en/listing/1607436850/third-eye-activation-spiritual-awakening

Image Transformations

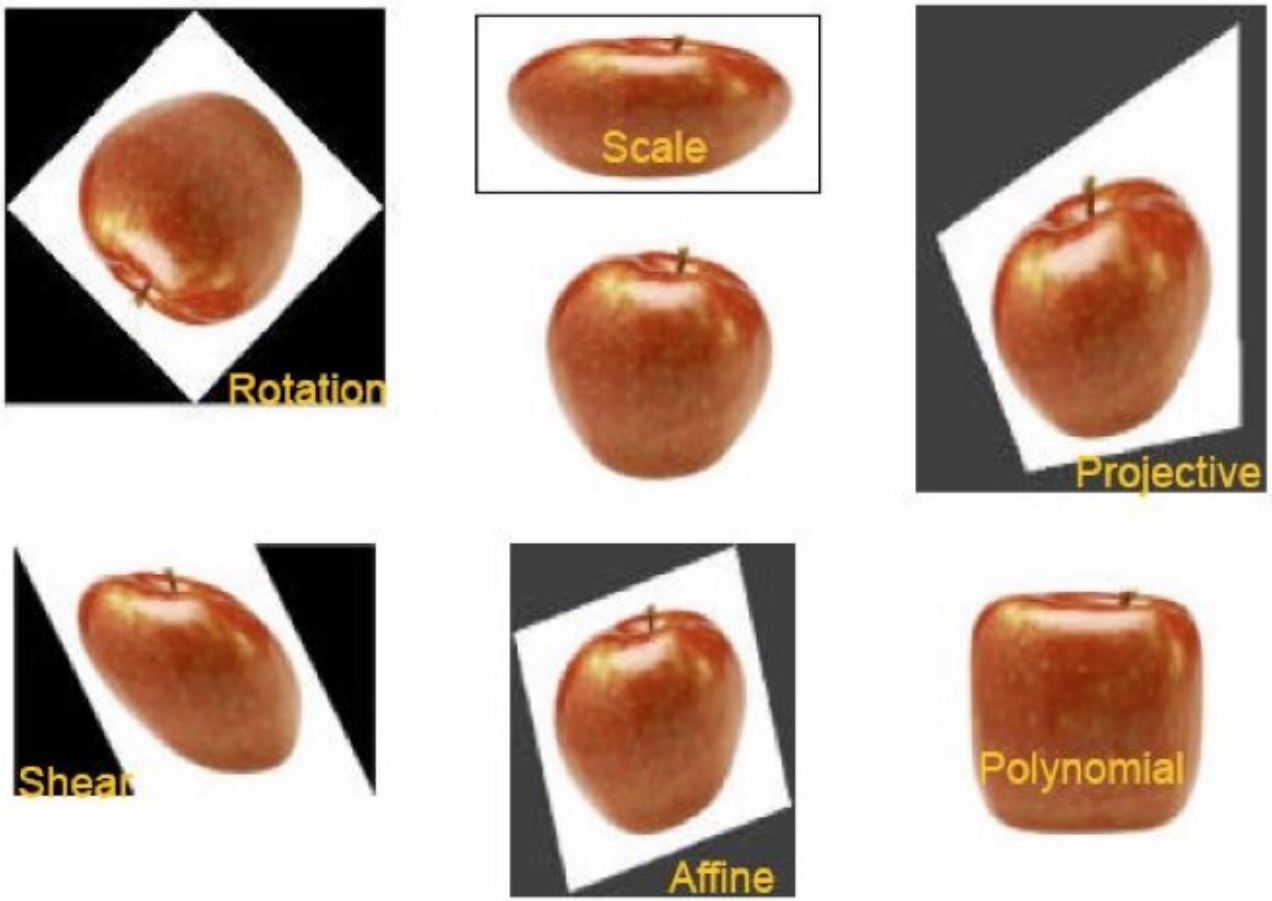
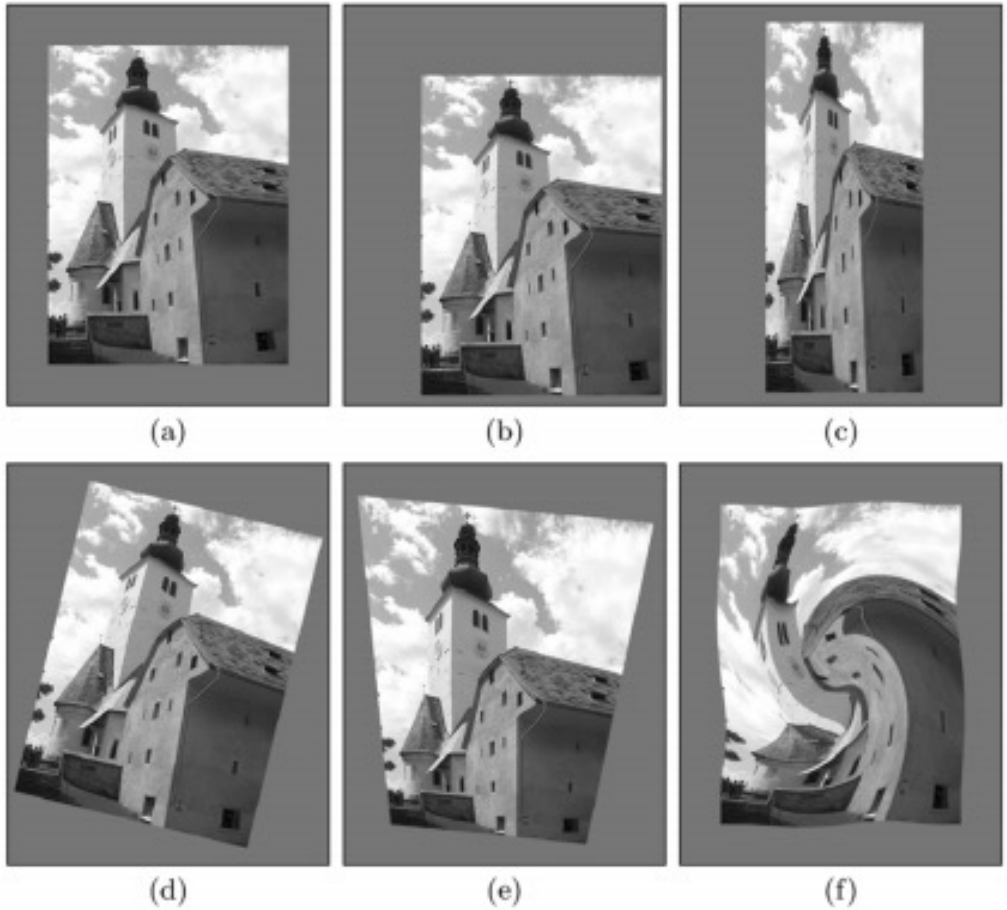


Image Transformations



Applications

- Align images
- Correct effects for camera orientation

- Correct images for lens distortion
- Image morphing

- Create interesting image effects

Geometric operations

- Geometric operation transforms image I to new image I' by modifying coordinates of image pixels:

$$I(x, y) \rightarrow I'(x', y')$$

- Intensity value (x, y) moved to a new position (x', y')

Example: Translation geometric operation moves value at (x,y) to (x + d_x, y + d_y)

Simple mappings (Translation)

- Shift by a vector (d_x, d_y)

$$T_x : x' = x + d_x$$
$$T_y : y' = y + d_y$$

or

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} d_x \\ d_y \end{pmatrix}$$

Simple mappings (Scaling)

- Contracting or Stretching along x or y axis by a factor of s_x or s_y

$$T_x : x' = s_x \cdot x$$
$$T_y : y' = s_y \cdot y$$

or

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} s_x & 0 \\ 0 & s_y \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$$

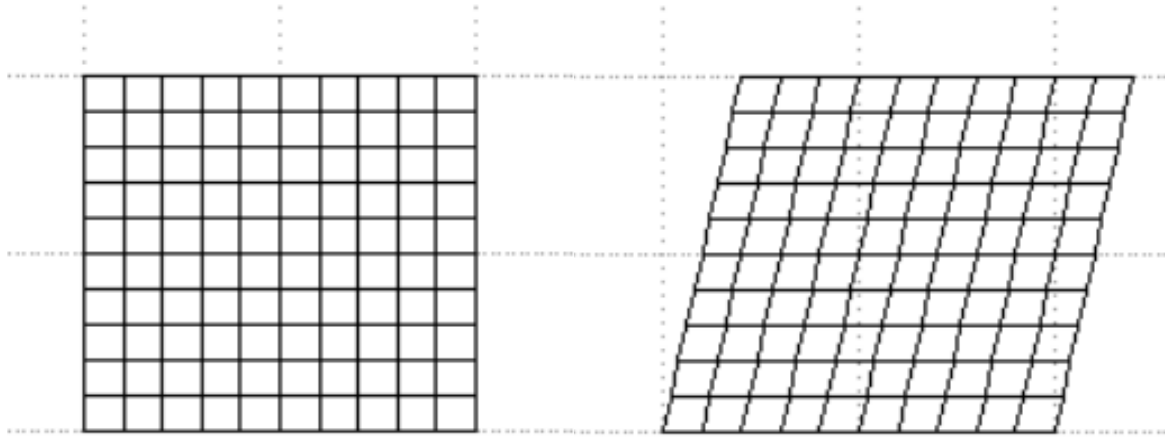
Simple mappings (Shearing)

- Along x and y axis by factor b_x and b_y

$$T_x : x' = x + b_x \cdot y$$
$$T_y : y' = y + b_y \cdot x$$

or

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & b_x \\ b_y & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$$



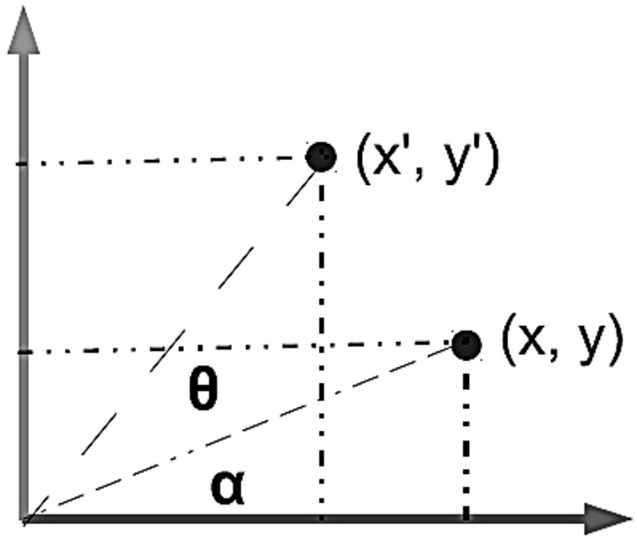
$x' = x + 0.2y$

Simple mappings (Rotation)

- 2D Rotation

Rotate about origin CCW by θ.

$$x' = x\cos\theta - y\sin\theta,$$
$$y' = x\sin\theta + y\cos\theta.$$



Matrix notation: $P' = R\,P$

$$\begin{bmatrix} x \\ y \end{bmatrix}' = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Simple mappings (Rotation)

$$T_x : x' = x \cdot \cos\theta \; - y \cdot \sin\theta$$
$$T_y : y' = x \cdot \sin\theta \; + y \cdot \cos\theta$$

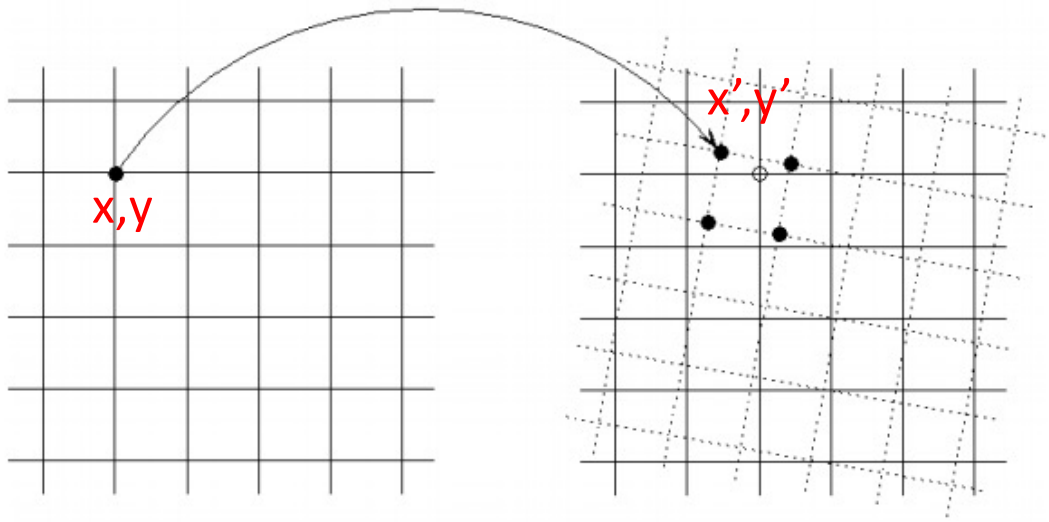
$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$$

Image flipping and rotation by 90 degrees

- Basic idea: look up a transformed pixel address instead of the current one
- To flip an image upside down (vertical flip)
  - At pixel location (x, y), look up the color at location $(m - x, y)$
- For horizontal flip
  - At pixel location (x, y), look up the color at location $(x, n - y)$

Image Warping

- Forward mapping (iterate over source image)

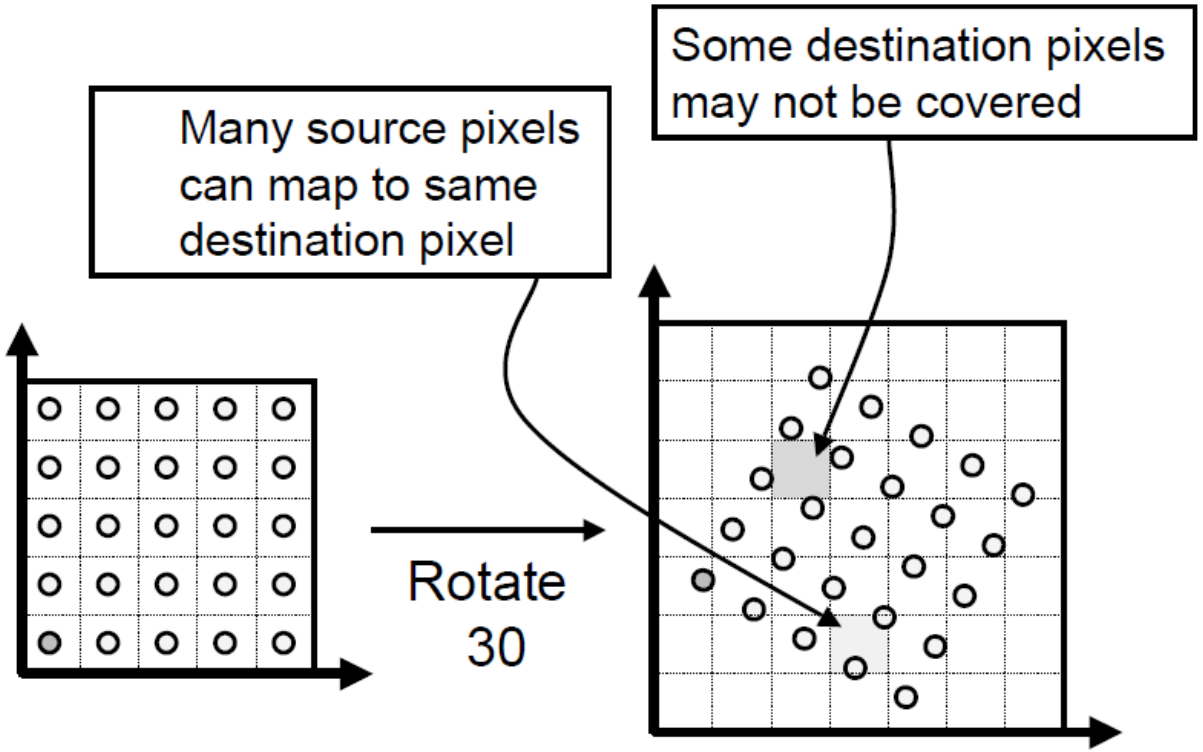



```
for (int x=0; x<W; x++)
    for(int y=0; y<H; y++)
        float x' = f_x(x, y);
        float y' = f_y(x, y);
        dst(x',y') = source(x,y);
    }
}
```

Transformed points x', y' may not fall on exact grid!!

Image Warping

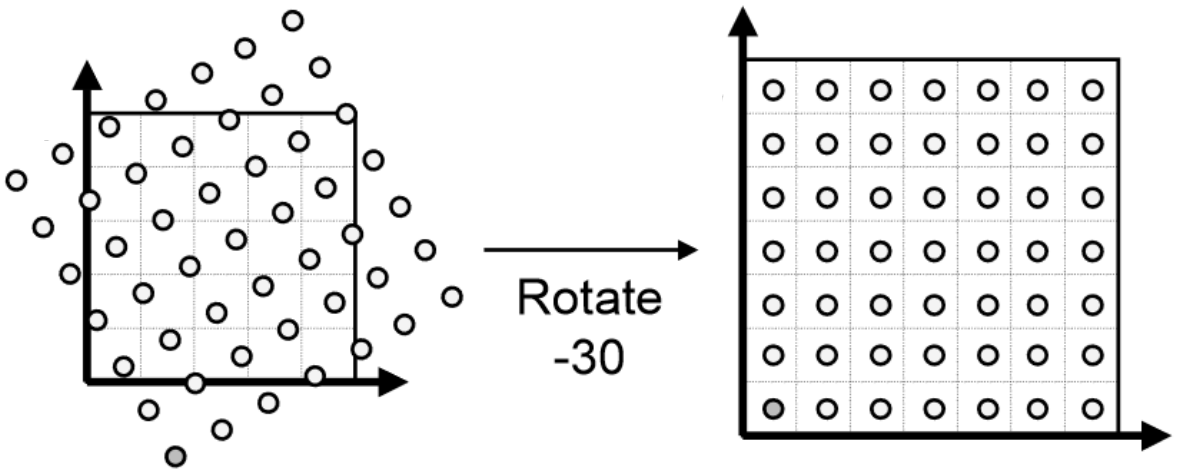
- Forward mapping (iterate over source image)



Courtesy: Thomas Funkhouser

Image Warping

- Reverse mapping (iterate over destination image)

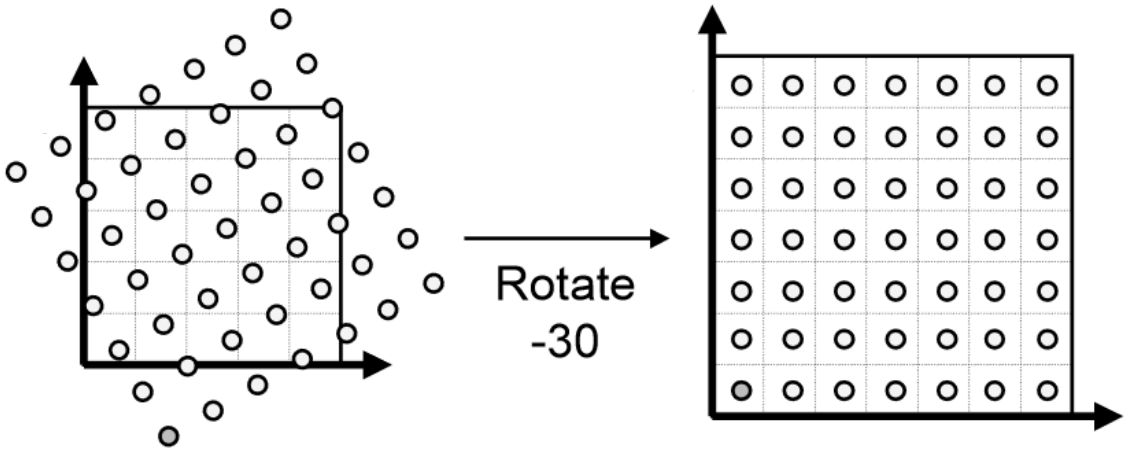



```
for (int x'=0; x'<W; x'++)
        for(int y'=0; y'<H; y'++)
                float x = f_x^{-1}(x', y');
                float y = f_y^{-1}(x', y');
                dst(x',y') = source(x,y);
        }
}
```

Courtesy: Thomas Funkhouser

Image Warping

- Reverse mapping (iterate over destination image)

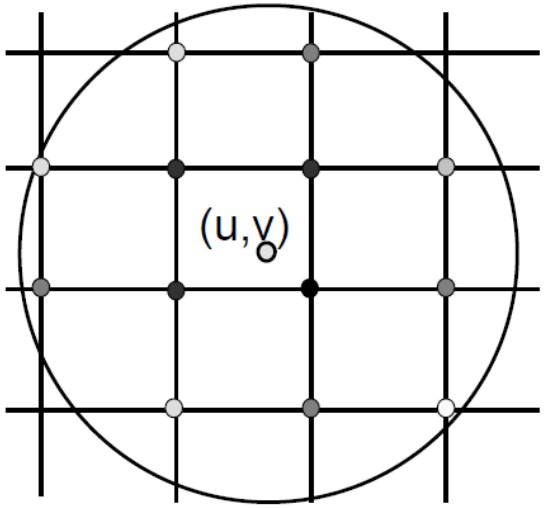



```
for (int x'=0; x'<W; x'++)
    for(int y'=0; y'<H; y'++)
        float x = f_x^{-1}(x', y');
        float y = f_y^{-1}(x', y');
        dst(x',y') = resample_src(x,y,w);
    }
}
```

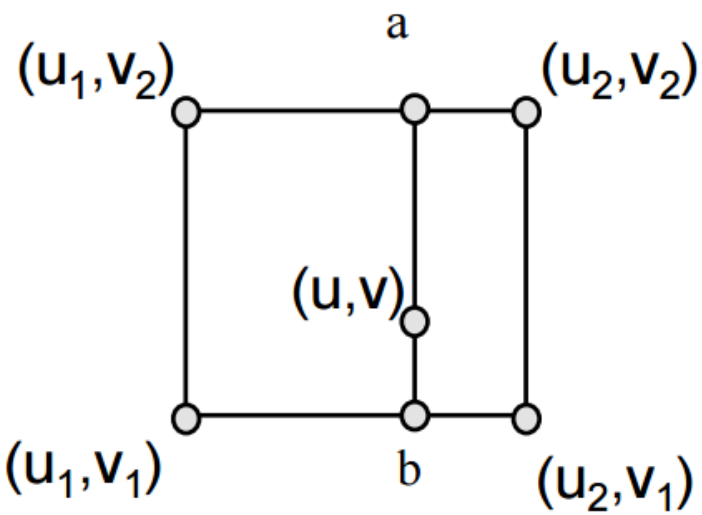
Courtesy: Thomas Funkhouser

Interpolation

- Reverse mapping (may not be exact integer)



Gaussian Interpolation



Bilinear Interpolation

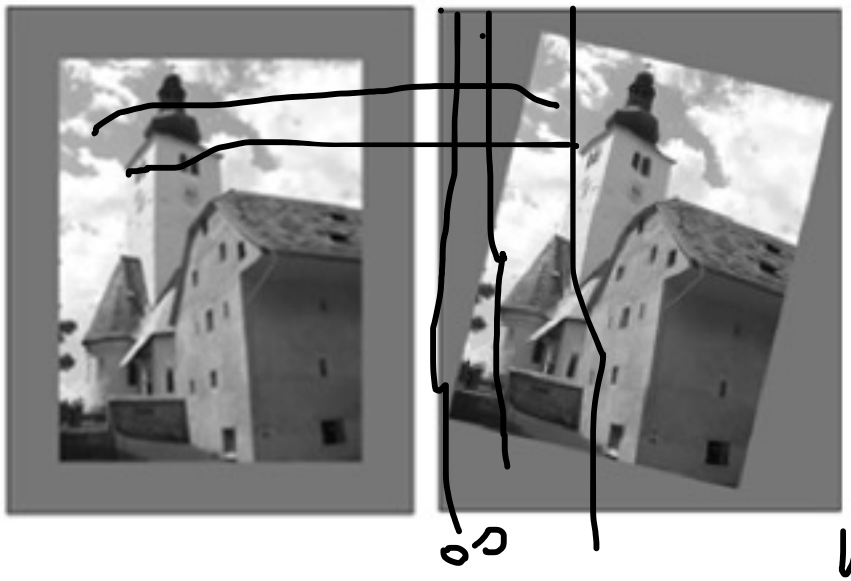
Courtesy: Thomas Funkhouser

Image Warping (Example Scaling)

$$T_x : x' = s_x \cdot x$$
$$T_y : y' = s_y \cdot y$$

```
for (int x'=0; x'<W; x'++)
        for(int y'=0; y'<H; y'++)
                float x = x'/s_x;
                float y = y'/s_y;
                dst(x',y') = resample_src(x,y,w);
        }
}
```

Image Warping (Example Rotation)



```
for (int x'=0; x'<W; x'++)
    for(int y'=0; y'<H; y'++)
        float x = x'cos(α) + y'sin(α);
        float y = −x'sin(α) + y'cos(α);
        dst(x',y') = resample_src(x,y,w);
    }
}
```

W x H

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$$

Transformation categories

- Planar
  - Rigid
  - Similarity
  - Affine
  - Projective

	Rotate	Translate	Scale	Skew/shear
Rigid	√	√	-	-
Similarity	√	√	√	-
Affine	√	√	√	√
Projective	√	√	√	√

Transformation categories

- Rigid (Isometric) → preserves length, angle, area

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

3 DOF (θ, t_x, t_y)

	Rotate	Translate	Scale	Skew/shear
Rigid	√	√	-	-
Similarity	√	√	√	-
Affine	√	√	√	√
Projective	√	√	√	√

(x, y) $\begin{bmatrix} x \\ y \end{bmatrix}_{2\times 1}$

Cartesian

$\begin{bmatrix} x \\ y \\ w=1 \end{bmatrix}_{3\times 1}$

homogeneous

$x = x/(w=1)$

$\begin{bmatrix} p \\ q \\ w \end{bmatrix}$

$p' = p/w$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} d_x \\ d_y \end{pmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \boxed{\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}} & \begin{matrix} d_x \\ d_y \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

3x3 3x

Transformation categories

- Similarity → parallel lines, angle, ratio between any two points

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

4 DOF (θ, t_x, t_y,s)

	Rotate	Translate	Scale	Skew/shear
Rigid	√	√	-	-
Similarity	√	√	√	-
Affine	√	√	√	√
Projective	√	√	√	√

Transformation categories

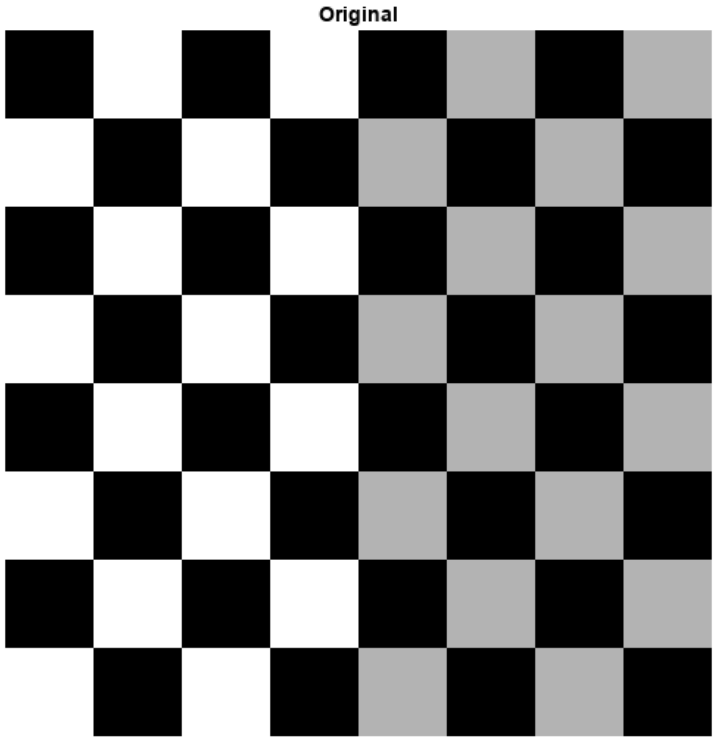
- Affine → collinear, parallel lines, ratio between any two points on a line

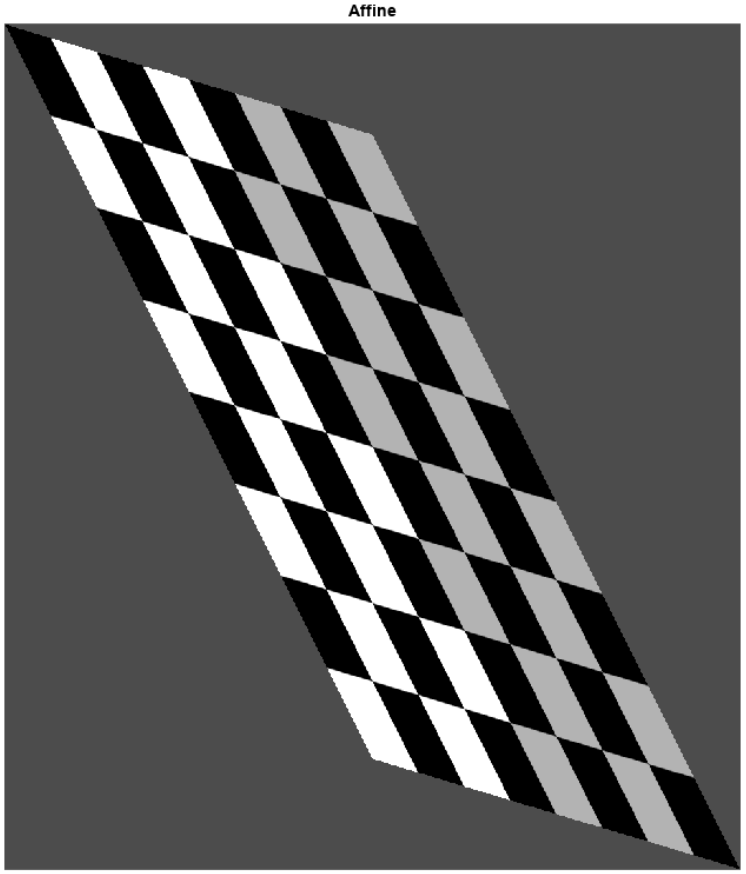
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

6 DOF ($a_{11}, a_{12}, a_{21}, a_{22}, t_x, t_y$)

	Rotate	Translate	Scale	Skew/shear
Rigid	√	√	-	-
Similarity	√	√	√	-
Affine	√	√	√	√
Projective	√	√	√	√

Affine Transformation





Transformation categories

- Projective → collinear, parallel lines not parallel

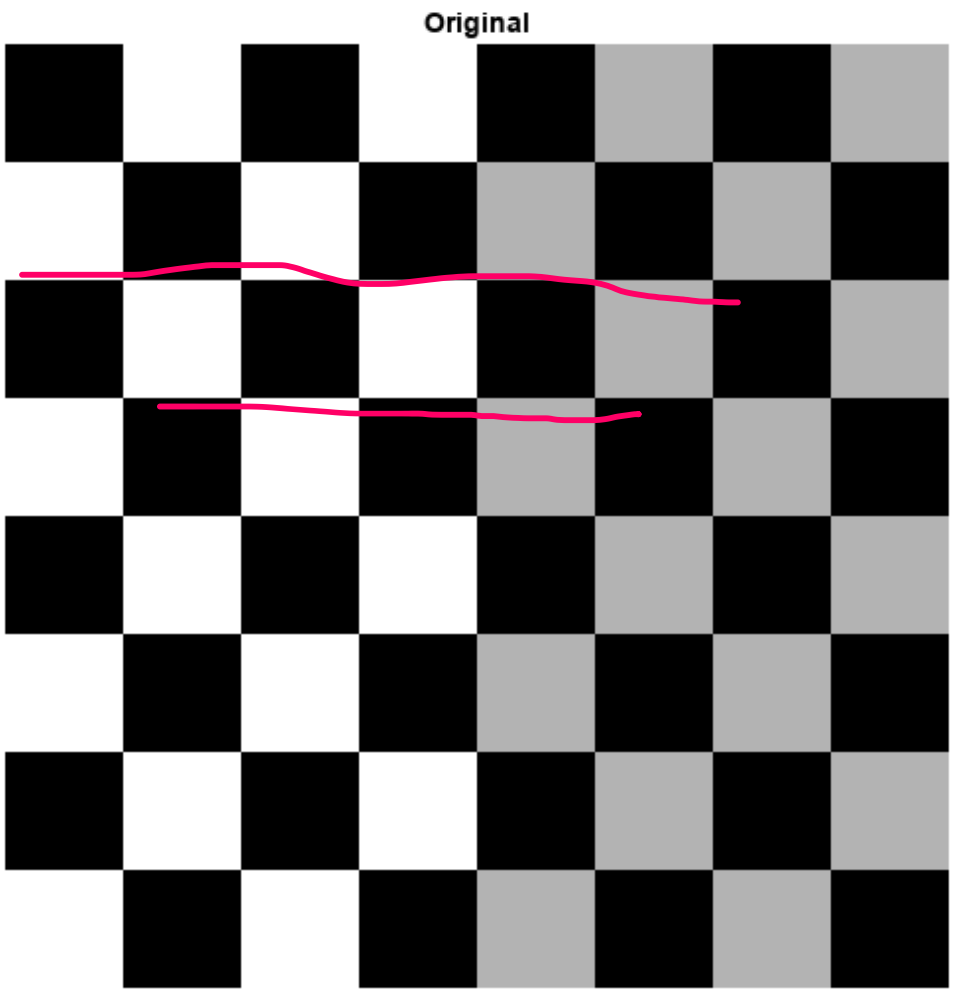
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

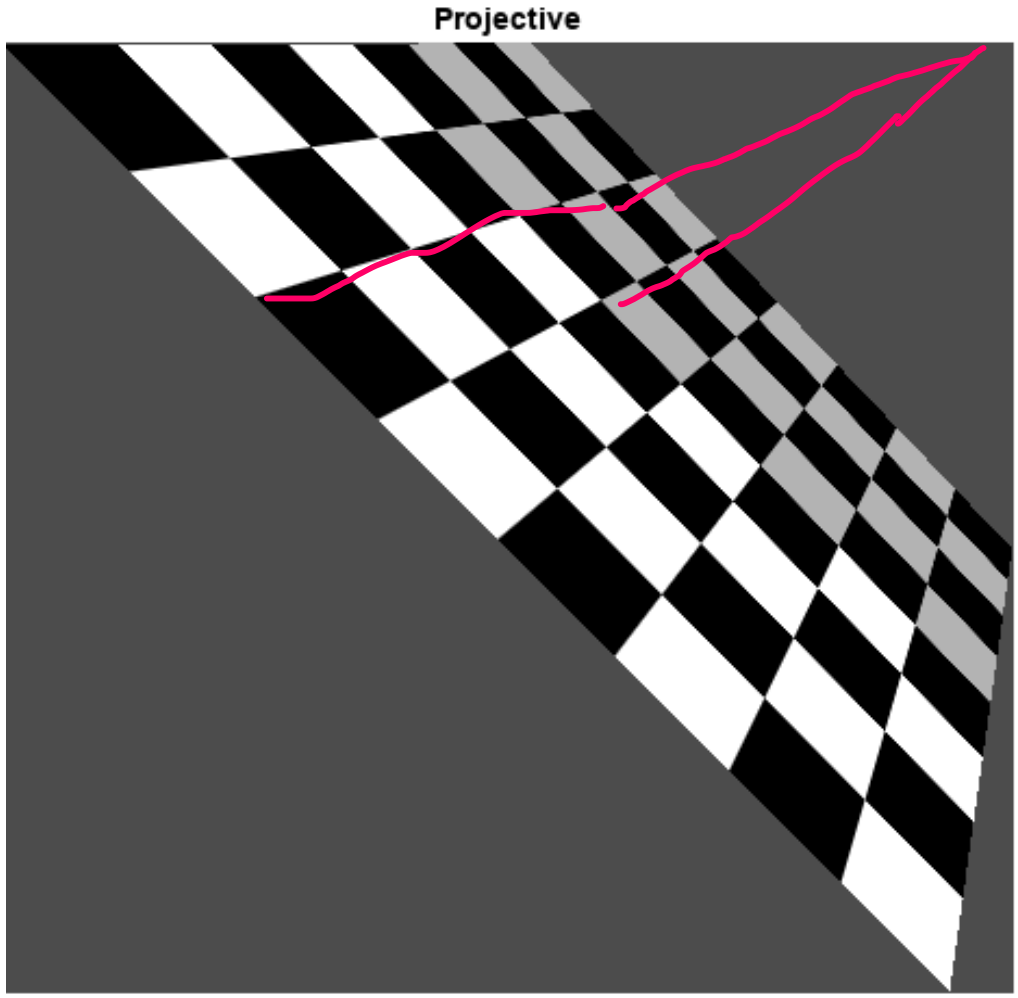
8 DOF

$$\begin{bmatrix} a_{11} & & a_{33} \\ a_{33} & & \\ & & 1 \end{bmatrix}$$

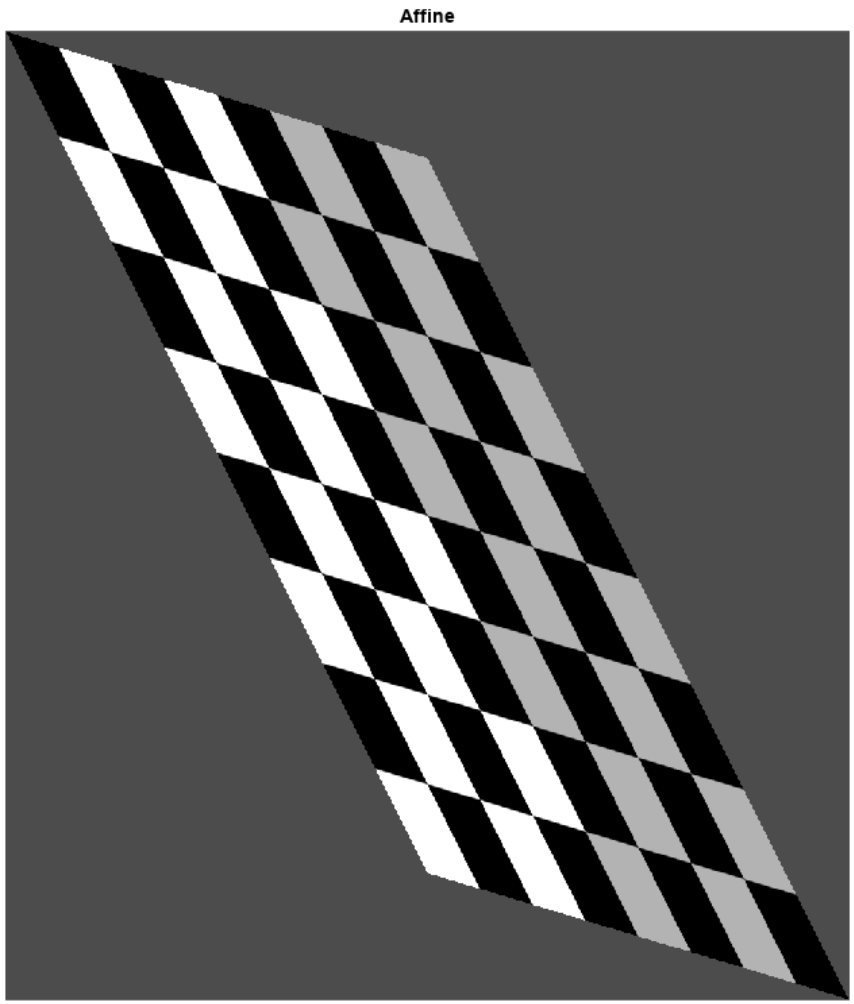
	Rotate	Translate	Scale	Skew/shear
Rigid	√	√	-	-
Similarity	√	√	√	-
Affine	√	√	√	√
Projective	√	√	√	√

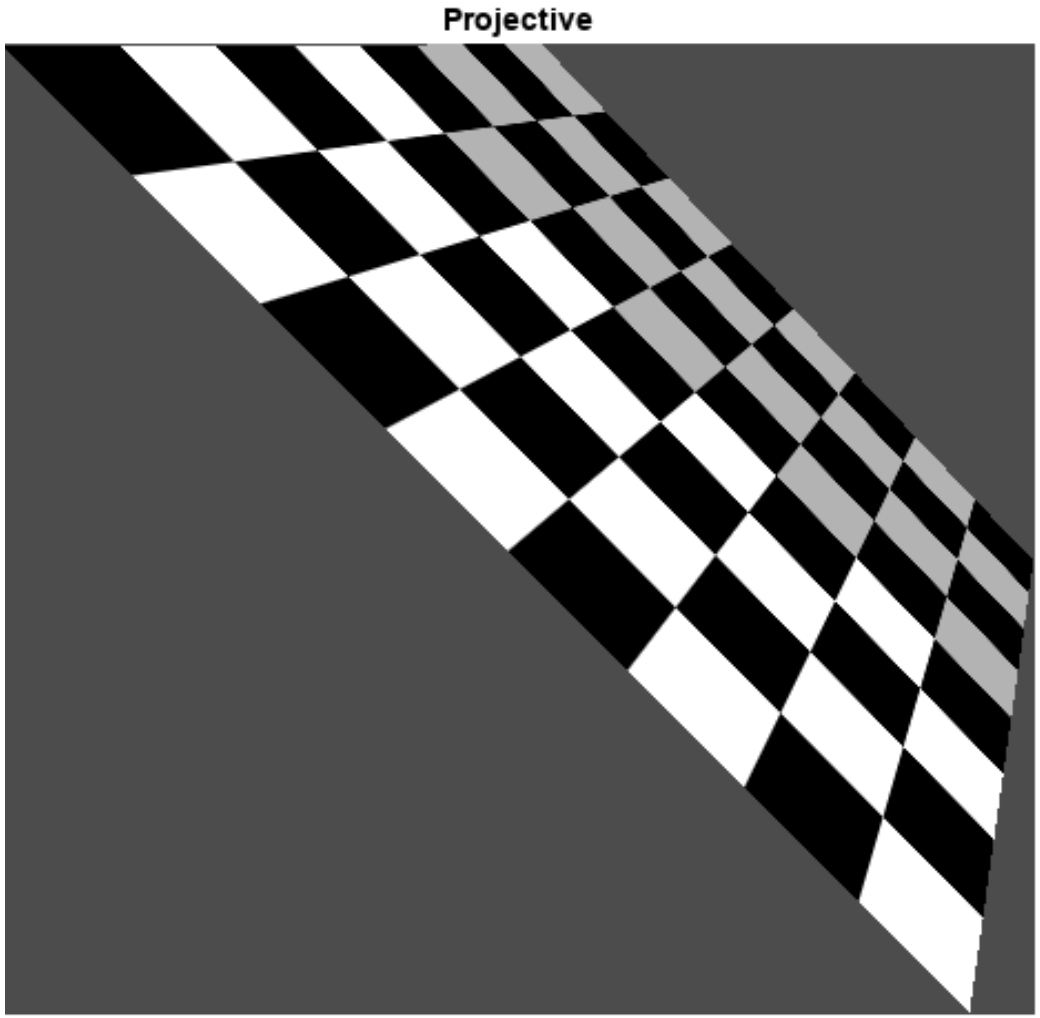
Projective Transformation





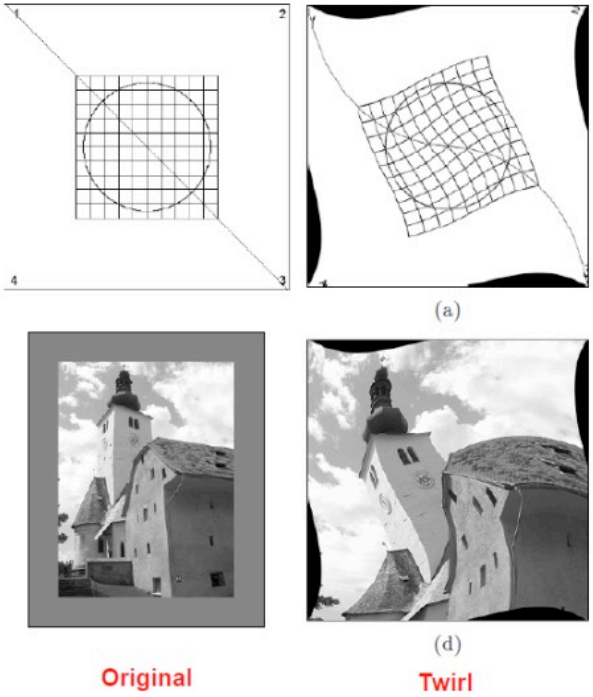
Affine Vs Projective Transformation





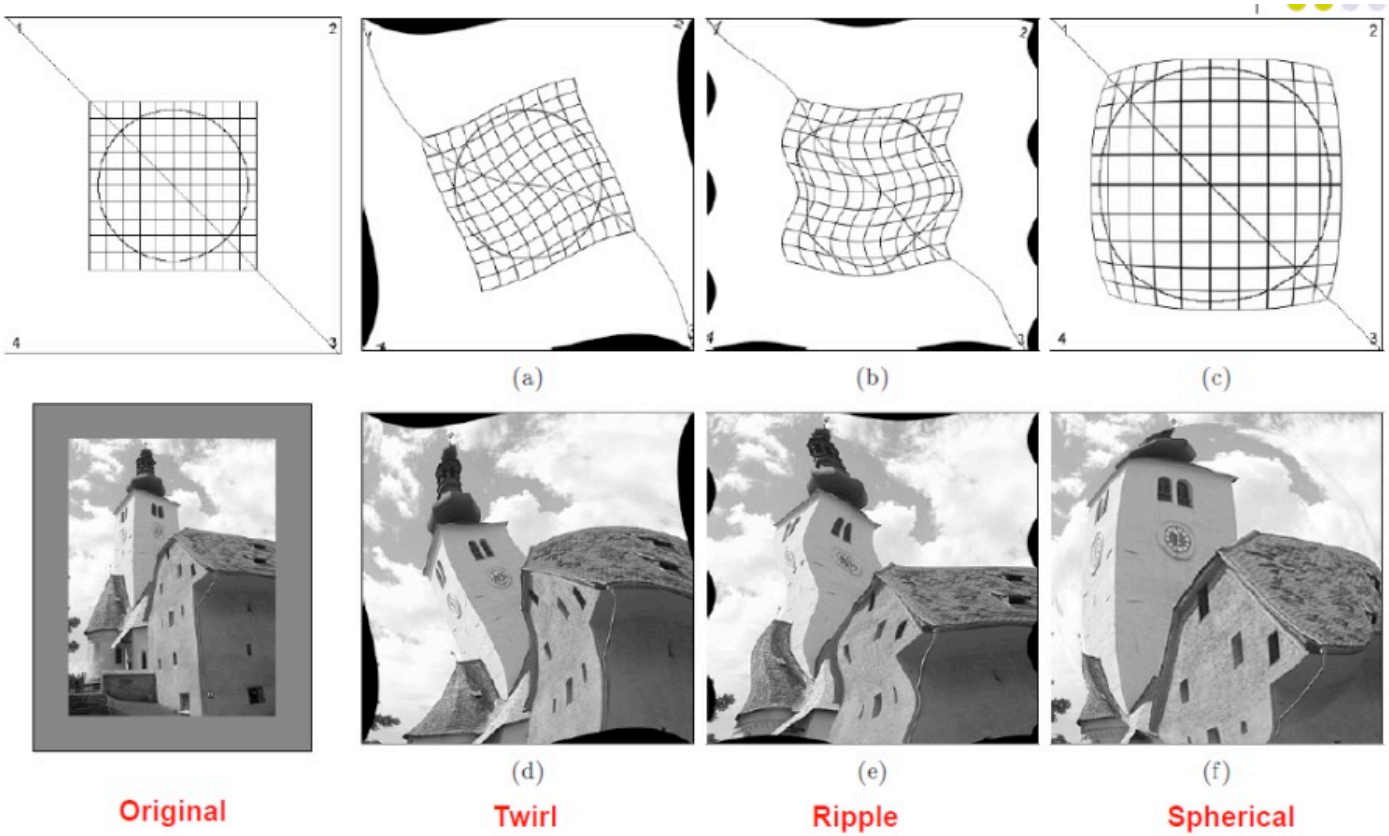
Transformation categories

- Non-linear
  - Curved → lines do not map to lines, shapes deform (expressed as polynomials)



Courtesy: Emmanuel Agu

Non Linear image warps



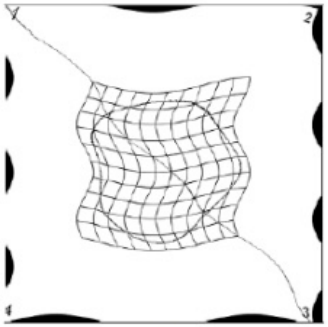
Courtesy: Emmanuel Agu

Ripple

- Wavelike displace along both x and y directions

$$T_x^{-1}: \quad x = x' + a_x \cdot \sin\left(\frac{2\pi \cdot y'}{\tau_x}\right),$$
$$T_y^{-1}: \quad y = y' + a_y \cdot \sin\left(\frac{2\pi \cdot x'}{\tau_y}\right).$$

Sample values of parameter: $a_x = 10, a_y = 15,$
$\tau_x = 120, \tau_y = 150$



(b)

(e)

Courtesy: Emmanuel Agu

Twirl

- Rotation about centre or some anchor point
  - Increasingly rotate as radial distance r from centre increases
  - Image unchanged after r_{max}

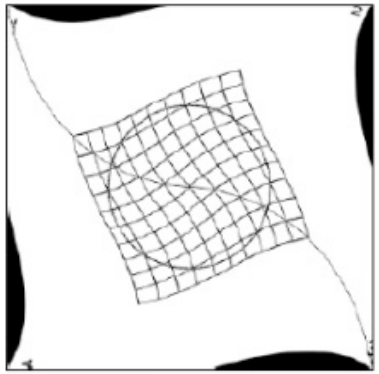
$$T_x^{-1}: \quad x = \begin{cases} x_c + r \cdot \cos(\beta) & \text{for } r \leq r_{\max} \\ x' & \text{for } r > r_{\max}, \end{cases}$$

$$T_y^{-1}: \quad y = \begin{cases} y_c + r \cdot \sin(\beta) & \text{for } r \leq r_{\max} \\ y' & \text{for } r > r_{\max}, \end{cases}$$

with

$$d_x = x' - x_c, \qquad r = \sqrt{d_x^2 + d_y^2},$$

$$d_y = y' - y_c, \qquad \beta = \mathrm{Arctan}(d_y, d_x) + \alpha \cdot \left(\frac{r_{\max} - r}{r_{\max}}\right).$$



(a)

Courtesy: Emmanuel Agu

Spherical Transformation

- Imitates viewing image through a lens placed over image
  - Increasingly rotate as radial distance r from center increases
  - Lens center (x_c, y_c), radius (r_{max}), refractive index (ρ)

Skew

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} ntan\theta \\ 0 \end{bmatrix}$$

$$T_x^{-1}: \quad x = x' - \begin{cases} z \cdot \tan(\beta_x) & \text{for } r \le r_{\max} \\ 0 & \text{for } r > r_{\max}, \end{cases}$$

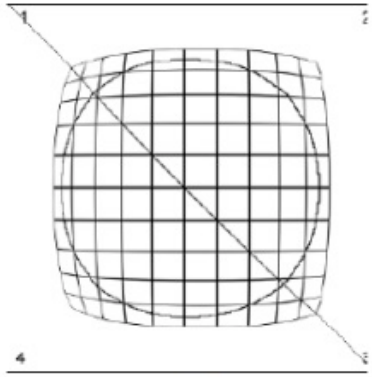
$$T_y^{-1}: \quad y = y' - \begin{cases} z \cdot \tan(\beta_y) & \text{for } r \le r_{\max} \\ 0 & \text{for } r > r_{\max}, \end{cases}$$

$$d_x = x' - x_c, \qquad r = \sqrt{d_x^2 + d_y^2},$$

$$d_y = y' - y_c, \qquad z = \sqrt{r_{\max}^2 - r^2},$$

$$\beta_x = \left(1 - \frac{1}{\rho}\right) \cdot \sin^{-1}\left(\frac{d_x}{\sqrt{(d_x^2 + z^2)}}\right),$$

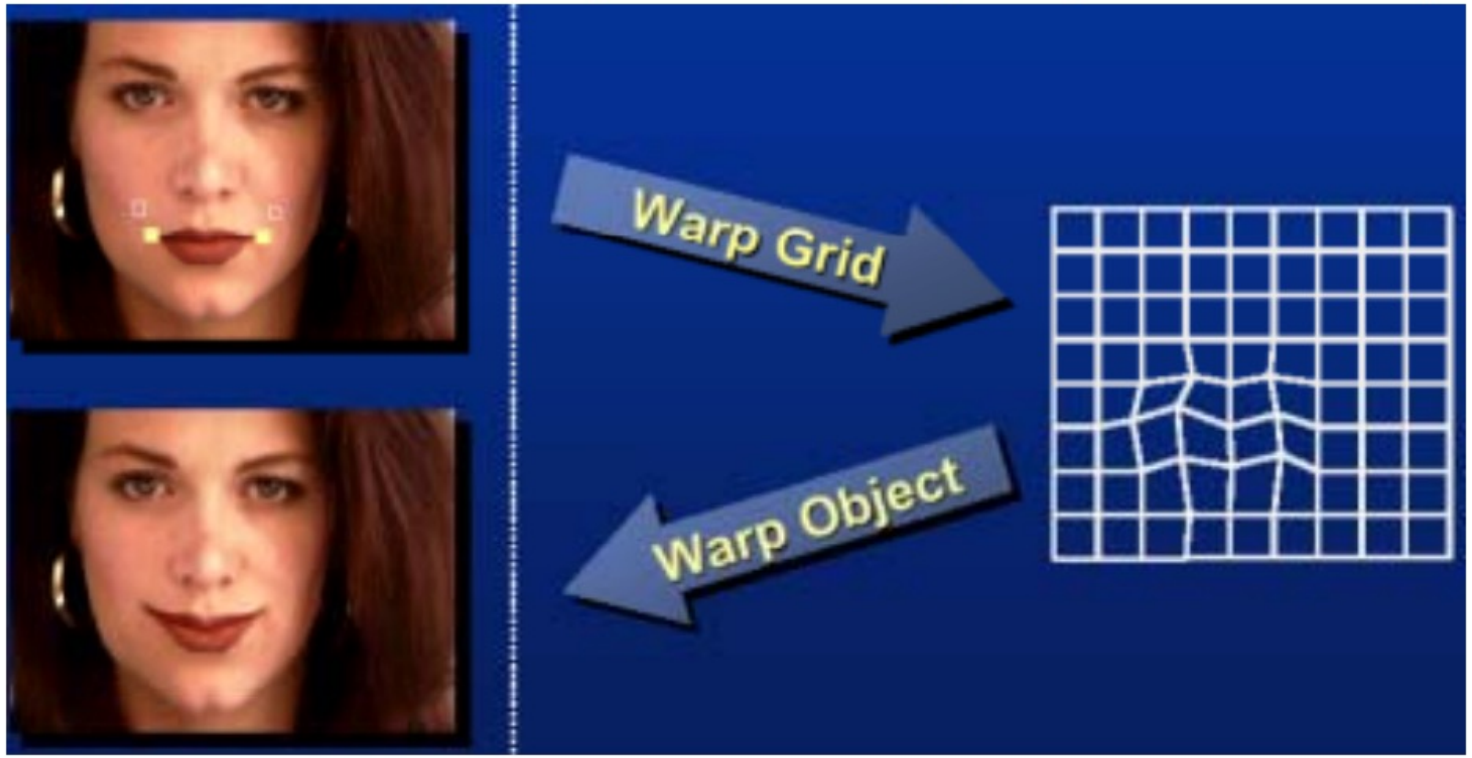
$$\beta_y = \left(1 - \frac{1}{\rho}\right) \cdot \sin^{-1}\left(\frac{d_y}{\sqrt{(d_y^2 + z^2)}}\right).$$



(c)

Courtesy: Emmanuel Agu

More specific grid based warping



Automatic Cinemagraph Portraits

EGSR 2013

Jiamin Bai[1] Aseem Agarwala[2] Maneesh Agrawala[1] Ravi Ramamoorthi[1]

UC Berkeley[1] Adobe[2]

Thank You